AC9M6ST01 • Year 6 Mathematics

Comparing Categorical and Numerical Data Sets

Interpret distributions using mode, shape, spread and context

We are learning to…

Students identify variable type, read displays accurately and compare centre, mode, range, clusters, gaps, skew and possible extremes without claiming causes unsupported by the data.

Represent | Reason | Calculate | Interpret

Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

interpret and compare data sets for ordinal and nominal categorical, discrete and continuous numerical variables using appropriate displays; compare distributions in terms of mode, range and shape

Compare two numerical distributions

feature	Class A	Class B
mode	**12**	**12 and 15**
range	**9**	5
shape	clustered 11–14, one high value	more even 11–16
variation	greater	smaller

One statistic does not describe a full distribution. Use several features and connect them to the measured context.

Match display to variable type

nominal categories	ordinal ratings	discrete numerical	continuous numerical	two distributions
column graph or table	ordered columns	dot plot or column graph	grouped display or line-related context	same scale and aligned categories

Comparisons require consistent scales and definitions. Avoid interpreting an association as a causal explanation.

Build and annotate the model

Represent comparing categorical and numerical data sets and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Ordinal categories treated as equal numerical intervals:** Order is meaningful, spacing may not be.
- **Range used as complete description:** Also discuss concentration and shape.
- **Different graph scales compared visually:** Read actual values.
- **Cause inferred from group difference:** Data comparison alone does not establish cause.

Quick check

- Classify a variable.
- Find mode/range.
- Describe shape.
- Compare two groups.
- Check graph scale.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.